

ATR First-Hour 50% Retracement TradingView Whitepaper

Explaining the chart strategy, the Pine Screener version, inputs, outputs, operational workflow, and platform-specific limitations

Bottom line. Use the strategy script on an individual stock chart for backtesting and trade simulation. Use the screener indicator in TradingView's Pine Screener to scan your watchlist for candidates. TradingView's own documentation now supports watchlist scanning with custom Pine indicators, but strategies still backtest one symbol at a time [TV1][TV2][TV3].

Executive Summary

The trading concept is straightforward: during the first hour of the U.S. regular session, look for a stock that moves at least **2× ATR** in one direction, then wait for a **50% retracement** of that impulse before looking for continuation in the original direction. The Market Rebellion article that closely matches the user-supplied short describes exactly that structure and recommends using ATR on the same timeframe as the trading chart [MR1].

The original chart strategy is best understood as a **single-symbol execution and backtesting tool**. TradingView's strategy engine still runs one symbol at a time, so it is ideal for validating entry logic, seeing simulated fills, and reviewing results in the Strategy Tester [TV3][TV4].

The companion screener indicator is best understood as a **candidate discovery tool**. TradingView's Pine Screener scans watchlists with indicators, not strategies, and uses a script's plots and alert conditions as columns and filters. The screener defaults to a 1-day interval, so this intraday setup requires changing the screener timeframe to 1-minute, 5-minute, or 15-minute depending on how you trade [TV1][TV2].

Strategy script vs. screener indicator

Dimension	Chart Strategy	Pine Screener Indicator	Best Use	Key Limitation
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Execution model	strategy()	indicator()	Use both together	Screener version cannot backtest
Symbols	One chart symbol at a time [TV3]	Scans a watchlist [TV1]	Screen first, test second	Strategy is not a portfolio scanner
Outputs	Orders, fills, P&L, lines, Strategy Tester	Plots, alert conditions, screener columns [TV1][TV2]	Find candidates	No simulated fills in Pine Screener
Timeframes	Any chart timeframe, but intraday recommended	Only supported screener intervals: 1, 5, 15, 30m; 1, 2, 4h; 1D, 1W, 1M [TV1]	Use 1m/5m for this setup	3m is not supported in Pine Screener
History window	Chart history / Deep Backtesting availability varies [TV3]	Last 500 bars only [TV1]	Good enough for same-day scans	Not for long multi-month statistical studies

1. What the chart strategy does

At the start of each regular U.S. session, the script records the day's opening price and resets all trade state. It then watches the first-hour scan window, which defaults to 09:30–10:30 America/New_York.

In “From session open” mode, the strategy measures the move away from the opening price. In the revised implementation included with this paper, wick mode uses the running first-hour high or low so the script can represent the strongest excursion from the open more faithfully than a single closing-price check. In “Single candle body” mode, the strategy instead evaluates an individual bar during the first hour.

Once price has moved at least the configured ATR multiple—2.0 by default—the script freezes the impulse, calculates the 50% retracement of that move, and places a limit order there. If it was an upside impulse, the strategy prepares a long; if it was a downside impulse, it prepares a short.

Risk management is explicit rather than inferred from the short. The stop is placed at the impulse base, the profit target is a configurable R-multiple from the entry level, and the script can optionally flatten at the end of the regular session. This stop/target framework is an implementation choice because the source material describes the entry setup clearly but is less specific about stock-share exit mechanics [MR1].

The strategy allows only one filled trade per session. That reduces duplicate signals and double-counting in backtests, but it can also miss second opportunities that occur later in the day.

Plain-English workflow

1. Wait for the U.S. regular session to open.
2. Watch the first hour for a move greater than the ATR threshold.
3. Measure 50% of that impulse.
4. Place a limit order at the retracement level.
5. Use the base of the impulse as the stop and a configurable R-multiple as the target.
6. Exit when the stop, target, or optional end-of-day rule is hit.

2. Inputs, required values, and how they affect results

The scripts are self-contained. They do not require external variables, files, symbols, or API keys. Every user-adjustable parameter is an input in the Pine settings pane. The table below lists the most important ones.

Input	Default	Required?	Purpose	Effect of changing it
ATR length	14	Optional	Length used by <code>ta.atr()</code>	Higher values smooth ATR and reduce trigger frequency; lower values make the threshold more reactive.
ATR trigger multiple	2.0	Optional	Minimum impulse size before a setup is armed	Higher values create fewer, stronger setups; lower values create more signals but more noise.
Retracement %	50	Optional	Entry pullback level	Lower values enter shallower pullbacks; higher values wait for deeper pullbacks and may reduce fills.
Trigger mode	From session open	Optional	Switches between cumulative session move and single-candle logic	Session-open mode is closer to the “move from the open” idea; single-candle mode is closer to the Market Rebellion write-up [MR1].
Use wick extreme	False	Optional	Measures impulses using highs/lows instead of closes	More setups will trigger because intrabar extremes count.
Longs / Shorts	True / True	Optional	Direction filters	Use one side only if you want to align with market bias or an external filter.
Time zone	America/New_York	Optional but important	Anchors session windows to the U.S. stock session	Changing it can shift the scan window and completely change which bars qualify.
Regular session	0930-1600	Optional	Defines the session reset logic	Changing it changes when the state resets and when the optional end-of-day behavior matters.
First-hour scan window	0930-1030	Optional	Defines when impulses may qualify	Widening it finds later moves; narrowing it makes the setup stricter.
Entry window	0930-1500	Optional	Defines how long the limit order can work	A shorter window reduces late-day fills; a longer window captures slower pullbacks.
Profit target (R multiple)	1.0	Optional	Sets target relative to entry risk	Higher values improve reward per winner but reduce hit rate.
Cancel unfilled entry after entry window	True	Strategy only	Cancels stale orders	Turning it off may allow much later fills that no longer represent the

				original setup.
Flat at regular-session close	True	Strategy only	Forces same-day exits	Turning it off allows overnight carry if the strategy is still in position.

Key internal variables

These are not user inputs, but they determine the script's behavior and explain how the code works under the hood.

Variable	Role	Why it matters
dayOpen	Opening price for the session	This is the anchor for "from session open" mode.
scanHigh / scanLow	Running extremes during the scan window	These make the cumulative first-hour version more faithful to the source setup.
impulseBase / impulseExtreme	Start and end of the qualifying move	They define the retracement, stop, and target.
retraceLevel	Planned entry price	This is the central output of the setup.
stopPrice / targetPrice	Risk and reward levels	These directly affect backtest results.
setupDir	Long or short state	This determines which order logic is active.
setupArmed	Whether a setup is waiting for pullback	Useful in both the strategy and the screener.
tradedToday / signalConsumed	One-signal-per-day guard	Reduces duplicate entries and repeat alerts.
setupBarIndex	Bar on which the setup was armed	Prevents same-bar order behavior and helps expose setup age in the screener.

3. When to run it and what outputs to expect

Use the chart strategy during intraday U.S. market hours on a standard candlestick chart. The setup makes the most sense on 1-minute, 5-minute, and 15-minute charts because the first-hour logic is sensitive to bar size and the source article explicitly recommends matching ATR to the chart timeframe [MR1].

When the strategy is added to a chart, expect four visual outputs: the session open line, the 50% retracement line, the stop line, and the target line. You will also see long or short setup markers when the 2× ATR impulse qualifies. If price later pulls back to the retracement level inside the entry window, the strategy places a simulated limit order there.

In the Strategy Tester, you should expect order statistics for that one symbol only. TradingView's broker emulator fills market orders on the next available tick by default, and strategy behavior can differ from live execution, especially on lower timeframes or thin names [TV4]. If you have access to Bar Magnifier, it can improve fill realism in backtests by using lower-timeframe data inside a larger bar [TV3].

For the screener version, expect no P&L or order simulation. Instead, expect a set of columns and optional alert conditions: Long Setup, Short Setup, Long Ready, Short Ready, Retrace Level, Stop Level, Target Level, ATR, Trigger xATR, and Setup Age Bars. These are the fields that the Pine Screener can display and filter [TV1][TV2].

4. Is it better as a chart strategy or a watchlist scanner?

Best answer: it works best as a two-part system. The strategy is better for a single stock chart, and the screener indicator is better for the watchlist.

Use the screener indicator first to search your watchlist for symbols that have already produced a qualifying first-hour impulse or have retraced to the 50% entry level.

Then open the candidate on a chart with the strategy applied to confirm the levels visually, inspect context, and review what the strategy would have done.

This division of labor matches TradingView's current architecture. Pine Screener scans watchlists with indicators and uses plots and alert conditions as filterable outputs, while strategies remain one-symbol backtesting tools [TV1][TV2][TV3].

The screener is especially useful because Pine Screener's plots still work for scanning even when hidden with `display.none`, which keeps the indicator clean on a chart while preserving the columns needed for watchlist scanning [TV2].

Recommended operational workflow

Phase	What to do	What to expect
Before 09:30 ET	Load a focused U.S. stock watchlist. Avoid scanning the whole market blindly.	Cleaner signals and faster triage.
09:30–10:30 ET	Run Pine Screener on 1m or 5m. Filter Long Setup > 0 or Short Setup > 0.	Names that have produced the qualifying ATR impulse.
After a setup appears	Watch Long Ready or Short Ready, or use screener alerts.	A candidate has retraced to the entry zone.
Trade decision	Open the symbol on a chart with the strategy applied.	Visual confirmation of entry, stop, and target.
Post-trade review	Check Strategy Tester results and screenshot examples.	You learn whether settings fit your stock universe.

5. How the Pine Screener version is designed

TradingView's Pine Screener can use one indicator per screen, requires at least one plot or alert condition, and automatically adds the first ten plots as screener columns. It also allows filtering by indicator plots and alert conditions [TV1].

That constraint shaped the screener script in this package. The script is an indicator, not a strategy. Its first ten `plot()` calls are intentionally reserved for screener use. The columns are hidden from the chart with `display.none`, but TradingView documents that hidden plots still work for watchlist scanning, alerts, exports, and the Pine Screener [TV2].

The screener version avoids unsupported Pine Screener inputs such as `input.timeframe()`, `input.symbol()`, and `input.time()`. It also fits within Pine Screener's supported interval list and does not use `request.*()` calls at all, which keeps it well inside the screener limits [TV1].

Because the Pine Screener calculates only the last 500 bars, it is appropriate for same-day setup discovery but not for large historical studies. For a first-hour intraday strategy, that is usually acceptable because the relevant information is concentrated in the current session anyway [TV1].

6. Limitations, assumptions, and why outcomes may change

The source setup clearly defines the entry logic, but not a universal stock-share stop/target model. The attached strategy therefore uses a practical implementation: stop at the impulse base and target as a configurable R-multiple. Different exit rules will materially change results.

The “From session open” interpretation and the “Single candle body” interpretation are not identical. Depending on whether you use cumulative session logic or single-bar logic, the same day can produce different signals.

Wick mode versus close mode matters a great deal. Wick mode will trigger more setups because transient highs and lows count even when the bar closes back inside the range.

The strategy freezes the impulse when the first qualifying trigger appears. A trader who re-measures the move after a later extension would get a different retracement price.

Pine strategy fills are simulated, not live. The broker emulator, next-tick assumptions, and end-of-session behavior can all make backtests look cleaner or worse than real trading [TV4].

Thin names and volatile gaps may create fills that are technically possible in the emulator but hard to execute manually.

7. Files included with this whitepaper

File	Purpose	Use it when
atr_first_hour_retracement_whitepaper.docx / .pdf	This explanation document	You want the logic, workflow, and platform notes.
atr_first_hour_retracement_strategy.pine	Backtesting / chart strategy	You want simulated orders and Strategy Tester results.
atr_first_hour_retracement_screener.pine	Watchlist scanner / Pine Screener indicator	You want to scan a watchlist for active setups.

References

[MR1] Market Rebellion, “Use This Simple ATR Strategy to Scalp & Day Trade Options,” Sep. 14, 2023. Used for the underlying >2 ATR / first-hour / 50% retracement setup description and the recommendation to match ATR to the chart timeframe.

[TV1] TradingView Help Center, “TradingView Pine Screener: key features and requirements,” accessed Mar. 7, 2026. Used for watchlist scanning, first ten plots, alert conditions, supported timeframes, and 500-bar limit.

[TV2] TradingView Pine Script® User Manual, “Visuals / Overview,” accessed Mar. 7, 2026. Used for the fact that plots can still be used externally for alerts and watchlist scanning regardless of display state, including display.none.

[TV3] TradingView Pine Script® FAQ, “Strategies,” accessed Mar. 7, 2026. Used for the statement that strategies run one symbol at a time and for Bar Magnifier guidance.

[TV4] TradingView Pine Script® User Manual, “Concepts / Strategies,” accessed Mar. 7, 2026. Used for next-tick order-fill behavior and the broker emulator framing.

[TV5] TradingView Pine Script® User Manual, “Concepts / Sessions,” accessed Mar. 7, 2026. Used for session variable terminology and session-string behavior.